

WILDFIRE SIMULATOR USER MANUAL

- A. When the simulator first opens, you will see (image 1 below):
- The terrain in the center of the screen,
 - Toggles and a drop down in the top left to select temperature, humidity, and wind values.
 - A compass square directly above the terrain (it will be a box with an "N" on it).
 - The cardinal direction this is showing is the direction you are facing. If the compass says "N" you are facing north.
 - You can rotate the terrain to the left or right using the "Q" and "E" keys on your keyboard. This is indicated in the top right of the screen.

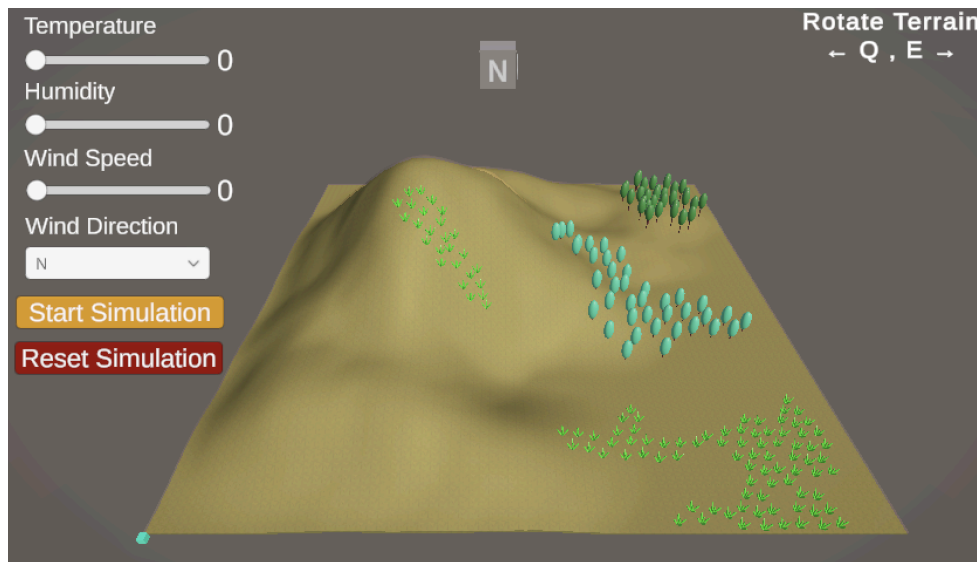


IMAGE 1: *The simulator opening scene*

- B. Running the Simulation:
- Click and drag the toggles for temperature, humidity, and wind speed to select your desired value.
 - Use the drop down to select the wind direction.
 - Click start
 - You will not be able to start a fire until you press start.
 - Parameters cannot be changed once you press start.
 - Click a spot on the terrain where you would like the fire to ignite. The fire will start.
 - Rotate the terrain to watch the fire spread.
 - You can ignite multiple fires if desired.
 - Press the "Reset" button to restart the simulation.
- C. Terrain Explanation (Image 2 below):
- There will be a blue box under your cursor showing you where on the map you are going to select.
 - For vegetation there is **grass**, **high trees** and **low trees**.

- i. Fire burns faster in the **grass**.
- ii. **High trees** = trees with no ladder fuel, so fire should slow down in these areas.
- iii. **Low trees** = trees with ladder fuel, so fire will burn faster through these.
- c. Burning ground is orange and extinguished ground is black.

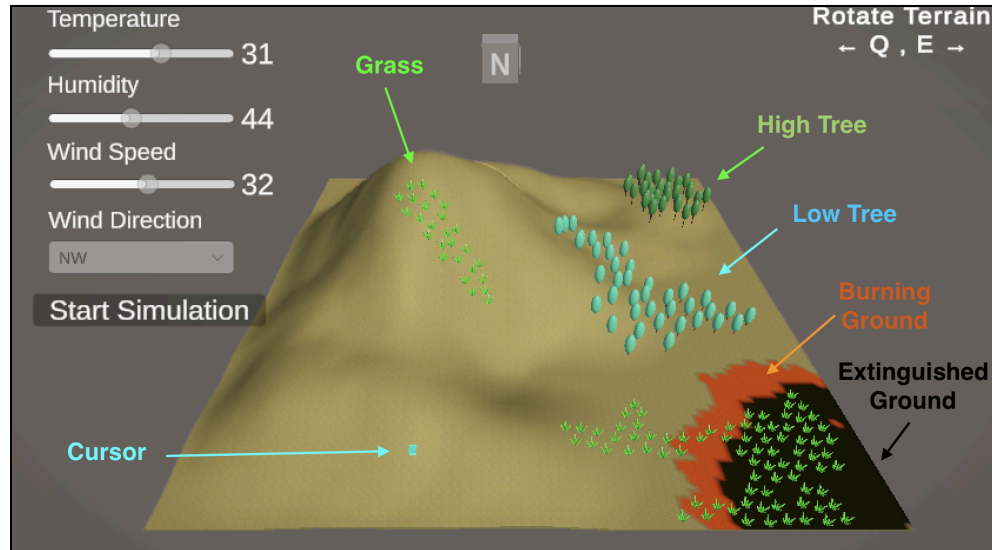


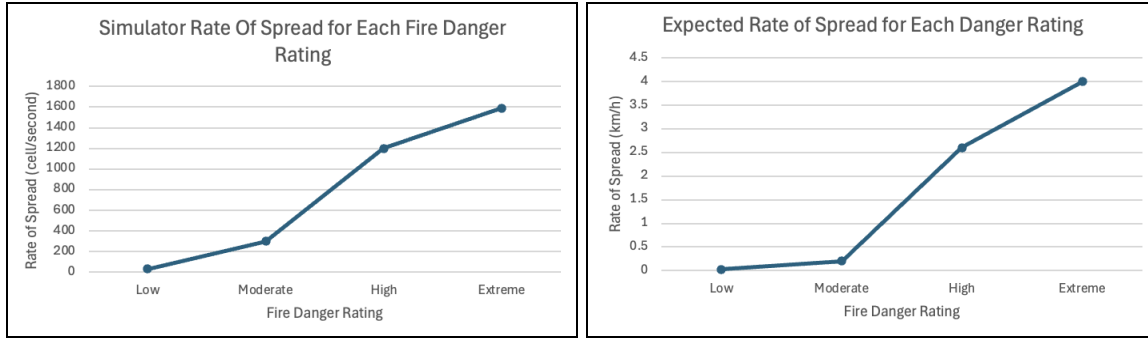
IMAGE 2: The simulator when running

D. Fire Behaviour in the simulator:

- a. The simulator considers the following variables:
 - i. Fuel
 - 1. Grass (burns quickly), low trees (burn quickly), high trees (burn slowly).
 - ii. Slope
 - 1. Fire burns faster upslope.
 - iii. Wind
 - 1. Fire burns faster downwind, relative to wind speed.
 - iv. Temperature:
 - 1. Fire burns faster in higher temperatures.
 - v. Humidity:
 - 1. Fire burns faster in lower humidities.
- b. You can adjust the parameters and where the fire is started to see these variables in action.

SIMULATOR RATE OF SPREAD (Compared to real world values)

The adjusted rate of spread was determined by first mapping the expected and calculated rates of spread, which presented a very similar pattern to one another (Graphs 2 and 3 below). Then a ratio was obtained to determine how much the simulator sped up rate of spread in order to adjust the rate of spread values obtained during testing trials (Table 9 below).



Graphs 2 and 3: Fire Danger Rating mapped to Rate of Spread for simulator and expected conditions. Note the similarities in slope trends.

TABLE 9: Calculation of the value to adjust the simulator values to get a realistic estimate of rate of spread.

Fire Danger Rating	Max Expected Rate of Spread (km/h)	Average Simulator Rate of Spread (km/h)	Ratio of expected / simulator rate of spread
Low	0.05 km/h	30 km/h	600
Moderate	0.5 km/h	300 km/h	600
High	2 km/h	1230 km/h	615
Extreme	3 km/h	1590 km/h	530
Average Ratio →			586.25 → Round to 600